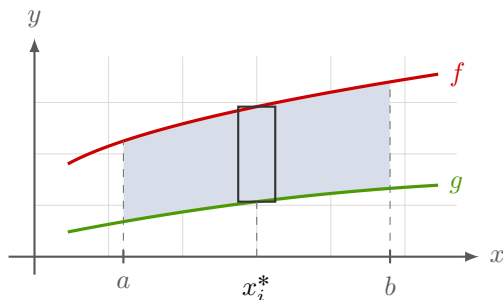


L4.1a Areas Between Curves

Work in your group. Check your answers against the Quarto tonight.

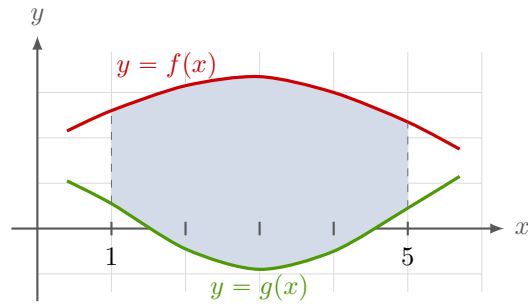
The area between two curves

1. One representative rectangle is drawn between the curves at the sample point x_i^* .



- (a) Write the **length** of that rectangle, and its **width**.
- (b) Write the sum that approximates the shaded area with n such rectangles.
- (c) Write the integral that sum becomes.
- (d) That integral holds only if f and g have one property in common. Name it.

2. No equations are given for either curve.



- (a) Write the integral that gives the shaded area.
- (b) Part of the region lies below the x -axis. Does that change what you wrote? One sentence.

Top minus bottom

3. The region bounded by $y = x^3$ and $y = x^2$, and on the sides by $x = 1$ and $x = 2$. Write the integrand, **label its two terms Top and Bot**, then find the area.

4. The region bounded by $y = x^2$ and $y = \sqrt{x}$, and on the sides by $x = 0$ and $x = 1$.

Stop. Which curve is on top here? Settle it by testing a point inside the interval, not by looking at the two formulas.

- (a) Find the area.
- (b) Write your set-up a second time using y_T and y_B instead of the two function names.

Limits from the intersections

5. Find the area of the region enclosed by $y = x^2$ and $y = x + 6$. Nobody hands you the limits.

Stop. You will have two intersection points. Which coordinate of each one is a limit of integration?

Curves that trade places

6. The region between $y = x$ and $y = x^3$ on $[-1, 1]$.
- (a) Sketch both curves, shade the region, and mark every point where they cross.
 - (b) $\int_{-1}^1 (x - x^3) dx = 0$. In one sentence, why is 0 not the area?
 - (c) Write the correct set-up and evaluate it.
 - (d) If two curves crossed **three** times on $[a, b]$, how many integrals would the area take, and how would you know which order to subtract in on each one?

7. A student is asked for the area of the region enclosed by $y = x^2$ and $y = 2x$. They write

$$A = \int_0^2 (x^2 - 2x) dx = -\frac{4}{3}.$$

The limits are right. Name the mistake in one sentence, then write the correct set-up and evaluate it.